Package 'golemirt'

November 5, 2026

Title Interactive Item Response Theory Models and Estimation

Version 0.0.0.9000

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Description A 'shiny' application, built as a 'golem' package, for learning item response theory (IRT) with interactive 'plotly' charts. It draws the response curves of the Rasch and one- to four-parameter logistic models, the partial credit, generalized partial credit, graded response, rating scale and nominal response models, the probit response functions of Thurstonian IRT, and the response and information surfaces of an item that measures two abilities. It also shows how the estimates are found: a person's ability by maximum likelihood with the Newton-Raphson method, an item's parameters by maximum likelihood on log-likelihood surfaces, and a person's ability by expected and maximum a posteriori estimation. It brings together the 'modelsirt' and 'mleirt' apps, adapted from the 'irtDemo' package.

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URL https://sedzinfo.github.io/golemirt/, https://github.com/sedzinfo/golemirt

BugReports https://github.com/sedzinfo/golemirt/issues

Depends R (>= 4.1.0)

Imports bslib,
config,
fGarch,
golem,
grDevices,
plotly,
shiny,
stats

Suggests htmltools,
testthat (>= 3.0.0)

Config/testthat/edition 3

Encoding UTF-8

Config/roxygen2/version 8.1.1

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`eap_map` *EAP and MAP estimates of ability by quadrature*

Description

Bayesian estimation of a person's ability from a response pattern, at quadrature points X_r spread evenly from -4 to 4. With prior weights $A(X_r)$ and likelihood $L(X_r)$, the expected a posteriori estimate

$$EAP = \frac{\sum_r X_r L(X_r) A(X_r)}{\sum_r L(X_r) A(X_r)}$$

is the mean of the posterior distribution, and the maximum a posteriori estimate is the quadrature point where the posterior $L(X_r)A(X_r)$ is highest. The defaults replicate de Ayala (2009, p. 79).

Usage

```
eap_map(
  responses,
  b = c(-2.155, -0.245, 0.206, 0.984, 1.211),
  a = rep(1, length(b)),
  quadrature = 10,
  mean = 0,
  sd = 1,
  skew = 1
)
```

Arguments

`responses`	Ones and zeros, the responses to the items.
`b`	Difficulties of the items; the default is the five items of de Ayala (2009, Table 4.4).
`a`	Discriminations of the items: 1 for the Rasch model.
`quadrature`	Number of quadrature points.
`mean`, `sd`	Mean and standard deviation of the prior.
`skew`	Skewness parameter of the prior, `xi` in `fGarch::dsnorm()`.

Value

A list with `x`, the quadrature points; `prior`, the prior weights; `trace`, the probability of each observed response at each point (one column per item); `likelihood`; `posterior`, the product of prior and likelihood; `eap`, `psd` (the posterior standard deviation) and `map`; and `responses` and `prior_mean`, as given.

References

de Ayala, R. J. (2009). *The Theory and Practice of Item Response Theory*. New York: Guilford Press.

Examples

```
estimate <- eap_map(response_pattern(2))
c(EAP = estimate$eap, MAP = estimate$map)
```

`info_mirt` *Information surface of a two-dimensional dichotomous item*

Description

The item information at each point: the squared slope of the response surface of `prob_mirt()`, in the direction in which it rises fastest, divided by $P(1-P)$,

$$I(\theta_1, \theta_2) = \frac{(\partial P/\partial\theta_1)^2 + (\partial P/\partial\theta_2)^2}{P(1-P)}.$$

For the compensatory model this is the information of the four-parameter logistic model (Magis, 2013) along the item vector, $D^2A^2(P-c)^2(d-P)^2/((d-c)^2P(1-P))$, with $A = \sqrt{a_1^2 + a_2^2}$. Where P is exactly 0 or 1, or $c = d$, the surface is flat and the information is 0.

Usage

```
info_mirt(
  theta1,
  theta2,
  a = c(1, 1),
  b = c(0, 0),
  c = 0,
  d = 1,
  D = 1.702,
  compensatory = TRUE
)
```

Arguments

`theta1`, `theta2`	Numeric vectors of abilities on the two dimensions.
`a`	Discriminations on the two dimensions.
`b`	Locations on the two dimensions.
`c`	Lower asymptote.
`d`	Upper asymptote.
`D`	Scaling constant.
`compensatory`	`TRUE` for the compensatory model, `FALSE` for the non-compensatory one.

Value

A matrix with one row for each value of `theta1` and one column for each value of `theta2`.

References

Magis, D. (2013). A note on the item information function of the four-parameter logistic model. *Applied Psychological Measurement*, 37(4), 304-315.

Examples

```
theta <- seq(-4, 4, length.out = 5)
info_mirt(theta, theta)
```

item_vector *Item vector of a compensatory two-dimensional item*

Description

The arrow that represents a compensatory item in the ability plane. It points in the direction in which the probability rises fastest, at the angle $\arccos(a_1/A)$ to the θ_1 axis; its length is the multidimensional discrimination $A = \sqrt{a_1^2 + a_2^2}$; and it starts on the line where the probability is halfway between the asymptotes, at the signed distance $(a_1 b_1 + a_2 b_2)/A$, the multidimensional difficulty, from the origin.

Usage

```
item_vector(a = c(1, 1), b = c(0, 0))
```

Arguments

a	Discriminations on the two dimensions.
b	Locations on the two dimensions.

Value

A list with `start` and `end`, the coordinates of the two ends of the arrow, `discrimination`, the multidimensional discrimination A, and `difficulty`, the multidimensional difficulty.

Examples

```
item_vector(a = c(1, 1), b = c(0, 0))
item_vector(a = c(2, 0.5), b = c(1, -1))
```

loglik_rasch *Log-likelihood of a response pattern under the Rasch model*

Description

The log of the probability of the given responses to items with known difficulties, at each ability,

$$LL(\theta) = \sum_i \left[u_i(\theta - b_i) - \log(1 + \exp(\theta - b_i)) \right],$$

where u_i is 1 for a correct response and 0 for an incorrect one. Its maximum is the maximum likelihood estimate of ability, which `newton_raphson_rasch()` looks for.

Usage

```
loglik_rasch(theta, responses, b)
```

Arguments

theta	Numeric vector of abilities.
responses	Ones and zeros, the responses to the items.
b	Difficulties of the items.

Value

A numeric vector of log-likelihoods, one for each value of theta.

Examples

```
loglik_rasch(c(-1, 0, 1), response_pattern(2), b = c(-1.90, -0.60, -0.25, 0.30, 0.45))
```

---

loglik_surface *Log-likelihood surface of an item's discrimination and difficulty*

Description

The log-likelihood of the responses of people with known abilities to one item, for every combination of discrimination and difficulty on a grid, under the three-parameter logistic model with a known lower asymptote (the two-parameter logistic model when c = 0) and $D = 1$.

Usage

```
loglik_surface(
  responses,
  theta,
  a = seq(-3, 3, by = 0.1),
  b = seq(-3, 3, by = 0.1),
  c = 0
)
```

Arguments

responses	Ones and zeros, the responses of the people.
theta	Abilities of the people.
a	Grid of discriminations.
b	Grid of difficulties.
c	Lower asymptote assumed for the item.

Value

A matrix with one row for each value of a and one column for each value of b, which keeps the two grids as its attributes a and b.

Examples

```
set.seed(1)
theta <- seq(-3, 3, length.out = 50)
surface <- loglik_surface(simulate_responses(theta, a = 1, b = -0.5), theta)
surface_maximum(surface)
```

`newton_raphson_rasch` *Newton-Raphson steps towards the maximum likelihood estimate of ability*

Description

Computes the log-likelihood of a response pattern under the Rasch model on a fine grid of abilities, fits a smoothing spline to it, and takes Newton-Raphson steps $\theta_1 = \theta_0 - LL'(\theta_0)/LL''(\theta_0)$ with the derivatives of the spline, as the MLE Rasch tab of the app does. For a raw score of 0 or of all items the log-likelihood has no maximum, and the steps move away.

Usage

```
newton_raphson_rasch(
  responses,
  b = c(-1.9, -0.6, -0.25, 0.3, 0.45),
  start = 0,
  iterations = 1,
  theta = seq(-6, 6, by = 0.001)
)
```

Arguments

`responses`	Ones and zeros, the responses to the items.
`b`	Difficulties of the items; the default is the five items of de Ayala (2009, p. 22).
`start`	Starting value of the estimate.
`iterations`	Number of steps.
`theta`	Grid of abilities on which the log-likelihood is computed.

Value

A list with `theta` and `loglik`, the grid and its log-likelihoods, and `steps`, a data frame with one row per step: `iteration`, the number of the step; `start` and `end`, the estimates the step starts from and ends at; `loglik_start` and `loglik_end`, the log-likelihood there; `d1` and `d2`, the first and second derivatives at the start; `intercept` and `crossing`, where the tangent at the start crosses the vertical and horizontal axes; and `angle`, the acute angle in degrees between the tangent and the horizontal axis.

References

de Ayala, R. J. (2009). *The Theory and Practice of Item Response Theory*. New York: Guilford Press.

Examples

```
nr <- newton_raphson_rasch(response_pattern(2), iterations = 5)
nr$steps[, c("start", "end")]
```

`plot_category_curves` *Plot category response curves*

Description

Draws the probability of each response category against ability, as in the single-trait tabs of the Models menu: one coloured line per category with a marker at every ability, a gray line at a probability of 0.5, and a legend in which a click hides or shows a curve.

Usage

```
plot_category_curves(
  theta,
  probs,
  title = "",
  subtitle = NULL,
  ylab = "P(X=m|theta)"
)
```

Arguments

`theta`	Numeric vector of abilities.
`probs`	Matrix of probabilities, one row for each value of `theta` and one column for each category, as returned by `prob_gpcm()` and the other `prob_*()` functions; the column names label the categories.
`title`	Title of the plot.
`subtitle`	Optional note, shown on a second, smaller line of the title.
`ylab`	Title of the vertical axis.

Value

A `plotly` object.

Examples

```
theta <- seq(-6, 6, by = 0.1)
plot_category_curves(theta, prob_grm(theta), "Graded Response Model")
p <- prob_4pl(theta, c = 0.2)
plot_category_curves(theta, cbind("1" = p, "0" = 1 - p), "3PL Model", ylab = "P(x)")
```

plot_loglik_surface *Plot a log-likelihood surface*

Description

Draws a surface from `loglik_surface()` in 3-D, with a red dot at its highest point: the maximum likelihood estimates of discrimination and difficulty.

Usage

```
plot_loglik_surface(loglik, a = attr(loglik, "a"), b = attr(loglik, "b"))
```

Arguments

`loglik`	Matrix of log-likelihoods, one row for each value of `a` and one column for each value of `b`.
`a, b`	The grids of discriminations and difficulties of the surface; by default those that `loglik_surface()` keeps with it.

Value

A `plotly` object.

Examples

```
set.seed(1)
theta <- seq(-3, 3, length.out = 50)
plot_loglik_surface(loglik_surface(simulate_responses(theta, a = 1, b = -0.5), theta))
```

plot_loglik_surfaces *Plot log-likelihood surfaces for several lower asymptotes*

Description

Draws one surface from `loglik_surface()` for each assumed lower asymptote (guessing value), in different colours on one vertical scale, each with a dot at its highest point. The legend gives the guessing value of each colour, and a click on an entry hides or shows that surface.

Usage

```
plot_loglik_surfaces(
  surfaces,
  guesses = seq(0.05, 0.6, by = 0.05),
  a = attr(surfaces[[1]], "a"),
  b = attr(surfaces[[1]], "b")
)
```

Arguments

surfaces	List of log-likelihood matrices, one for each value of guesses, on the same grids.
guesses	The lower asymptotes assumed for the surfaces.
a, b	The grids of discriminations and difficulties of the surfaces; by default those that loglik_surface() keeps with them.

Value

A plotly object.

Examples

```
set.seed(1)
theta <- seq(-3, 3, length.out = 50)
grid <- seq(-3, 3, by = 0.15)
guesses <- c(0.1, 0.3, 0.5)
responses <- simulate_responses(theta, a = 1, b = -0.5, c = 0.4)
surfaces <- lapply(guesses, function(g) loglik_surface(responses, theta, grid, grid, c = g))
plot_loglik_surfaces(surfaces, guesses)
```

---

plot_mirt_contour *Plot the contours of a two-dimensional item*

Description

Draws the response surface of prob_mirt() seen from above, as labelled contour lines of equal probability, with the zero lines of both axes, a thick line at one probability (usually halfway between the asymptotes) and, optionally, the item vector as a blue arrow. Both axes have the same scale, so the angle of the arrow is true.

Usage

```
plot_mirt_contour(
  theta1,
  theta2,
  z,
  nlevels = 10,
  level = NULL,
  arrow = NULL,
  xlab = NULL,
  ylab = NULL
)
```

Arguments

theta1, theta2	Abilities on the two dimensions.
z	Matrix with one row for each value of theta1 and one column for each value of theta2.
nlevels	Approximate number of contour levels.
level	Probability of the thick line, or NULL for none; every piece of it is drawn, and none when the grid does not reach it.
arrow	Item vector from item_vector(), or NULL; it is drawn when it lies inside the plotted grid.
xlab, ylab	Titles of the axes; NULL labels the abilities with the Greek letter theta, as theta1 and theta2.

Value

A plotly object.

Examples

```
theta <- seq(-4, 4, length.out = 50)
plot_mirt_contour(theta, theta, prob_mirt(theta, theta), level = 0.5, arrow = item_vector())
```

---

plot_mirt_surface *Plot a surface over the ability plane*

Description

Draws a matrix from prob_mirt() or info_mirt() as an interactive 3-D surface, coloured from purple (low) to yellow (high), that can be rotated and zoomed with the mouse.

Usage

```
plot_mirt_surface(
  theta1,
  theta2,
  z,
  xlab = NULL,
  ylab = NULL,
  zlab = "P(Y=1)",
  zrange = c(0, 1),
  eye = list(x = 1.5, y = -1.5, z = 1.5)
)
```

Arguments

theta1, theta2	Abilities on the two dimensions.
z	Matrix with one row for each value of theta1 and one column for each value of theta2.
xlab, ylab, zlab	Titles of the axes; NULL labels the abilities with the Greek letter theta, as theta1 and theta2.
zrange	Range of the vertical axis, or NULL to fit the surface.
eye	Position of the camera, as in plotly's scene.camera.eye.

Value

A plotly object.

Examples

```
theta <- seq(-4, 4, length.out = 50)
plot_mirt_surface(theta, theta, prob_mirt(theta, theta))
plot_mirt_surface(theta, theta, info_mirt(theta, theta),
  zlab = "Information", zrange = NULL, eye = list(x = -1.5, y = 1.5, z = 1.5)
)
```

---

plot_newton_raphson *Plot a Newton-Raphson step*

Description

Draws the log-likelihood from newton_raphson_rasch() with its last step: the estimate the step starts from (red dot), the tangent to the log-likelihood there (blue line), the estimate the step ends at (green dot) and the triangle between the tangent and the horizontal axis, whose angle is zero at the maximum.

Usage

```
plot_newton_raphson(nr)
```

Arguments

nr	Result of newton_raphson_rasch().

Value

A plotly object.

Examples

```
plot_newton_raphson(newton_raphson_rasch(response_pattern(2), iterations = 2))
```

plot_prior *Plot the parts of an EAP and MAP estimation*

Description

Draw the parts of an estimation from `eap_map()`:

`plot_prior()` the prior weights at the quadrature points;

`plot_tracelines()` the probability of each observed response across ability, blue for items answered correctly and red for items answered incorrectly;

`plot_likelihood()` the likelihood, the product of those probabilities;

`plot_posterior()` the posterior, with lines and values for the EAP (blue) and the MAP (red);

`plot_eap_combined()` the prior (red), likelihood (blue) and posterior (black, rescaled to be comparable) in one plot, with dashed lines at the prior mean and the posterior mean.

Usage

```
plot_prior(estimate)

plot_tracelines(estimate)

plot_likelihood(estimate)

plot_posterior(estimate)

plot_eap_combined(estimate)
```

Arguments

`estimate`	Result of `eap_map()`.

Value

A `plotly` object.

Examples

```
estimate <- eap_map(response_pattern(2))
plot_prior(estimate)
plot_tracelines(estimate)
plot_likelihood(estimate)
plot_posterior(estimate)
plot_eap_combined(estimate)
```

plot_tirt *Plot a Thurstonian IRT response function*

Description

Draws a probability from `prob_tirt()` or its pair and intercept forms against the trait, with a marker at every value and a gray line at 0.5.

Usage

```
plot_tirt(eta, p)
```

Arguments

`eta`	Numeric vector of trait values.
`p`	Probabilities, one for each value of `eta`.

Value

A `plotly` object.

Examples

```
eta <- seq(-6, 6, by = 0.1)
plot_tirt(eta, prob_tirt(eta))
```

prior_weights *Prior weights at quadrature points*

Description

The density of a skew-normal prior (`fGarch::dsnorm()`, which is the normal density when `skew = 1`) at the quadrature points, rescaled to sum to 1.

Usage

```
prior_weights(x, mean = 0, sd = 1, skew = 1)
```

Arguments

`x`	Quadrature points.
`mean, sd`	Mean and standard deviation of the prior.
`skew`	Skewness parameter of the prior, `xi` in `fGarch::dsnorm()`.

Value

A numeric vector of weights, one for each value of `x`.

Examples

```
prior_weights(seq(-4, 4, length.out = 10))
```

`prob_4pl` *Four-parameter logistic item response function*

Description

The probability of a correct (or keyed) response to a dichotomous item,

$$P(X = 1 \mid \theta) = c + \frac{d - c}{1 + \exp(-Da(\theta - b))},$$

which is the Rasch model when $a = 1$, $c = 0$ and $d = 1$, the 2PL when $c = 0$ and $d = 1$, and the 3PL when $d = 1$.

Usage

```
prob_4pl(theta, a = 1, b = 0, c = 0, d = 1, D = 1)
```

Arguments

`theta`	Numeric vector of abilities.
`a`	Discrimination.
`b`	Difficulty (location).
`c`	Lower asymptote (guessing).
`d`	Upper asymptote (inattentiveness).
`D`	Scaling constant: 1 keeps the logistic metric, and 1.702 makes the curve almost coincide with the normal ogive.

Value

A numeric vector of probabilities, one for each value of `theta`.

Examples

```
prob_4pl(c(-2, 0, 2))
prob_4pl(c(-2, 0, 2), a = 1.5, b = 0.5, c = 0.2, D = 1.702)
```

prob_gpcm *Category response curves of the generalized partial credit model*

Description

The probabilities of the categories $1, \ldots, m+1$ of an item with m step parameters,

$$P(X = k \mid \theta) = \frac{\exp\left(\sum_{v=1}^{k-1} Da(\theta - \delta_v)\right)}{\sum_{h=1}^{m+1} \exp\left(\sum_{v=1}^{h-1} Da(\theta - \delta_v)\right)},$$

where an empty sum is 0. With $a = 1$ this is the partial credit model (Masters, 1982), otherwise the generalized partial credit model (Muraki, 1992). The curves of categories k and $k+1$ cross at $\theta = \delta_k$.

Usage

```
prob_gpcm(theta, a = 1, steps = c(-2, -0.75, 0.75, 2), D = 1)
```

Arguments

theta	Numeric vector of abilities.
a	Discrimination.
steps	Step parameters $\delta_1, \ldots, \delta_m$.
D	Scaling constant: 1 keeps the logistic metric, and 1.702 makes the curve almost coincide with the normal ogive.

Value

A matrix with one row for each value of theta and one column for each category, named 1 to m + 1.

References

Masters, G. N. (1982). A Rasch model for partial credit scoring. *Psychometrika*, 47(2), 149-174.

Muraki, E. (1992). A generalized partial credit model: Application of an EM algorithm. *Applied Psychological Measurement*, 16(2), 159-176.

Examples

```
prob_gpcm(c(-2, 0, 2))
prob_gpcm(c(-2, 0, 2), a = 1.5, steps = c(-1, 0, 1))
```

prob_grm *Category response curves of the graded response model*

Description

The probability of a response above each boundary between adjacent categories is $P_k^*(\theta) = 1/(1 + \exp(-Da(\theta - b_k)))$, and the probability of each category is the difference between successive boundary curves, with $P_0^* = 1$ and $P_{m+1}^* = 0$. The difficulties must increase, or some probabilities would be negative.

Usage

```
prob_grm(theta, a = 1.5, b = c(-2, -0.75, 0.75, 2), D = 1)
```

Arguments

theta	Numeric vector of abilities.
a	Discrimination.
b	Increasing difficulties of the boundaries, $b_1, \ldots, b_m$.
D	Scaling constant: 1 keeps the logistic metric, and 1.702 makes the curve almost coincide with the normal ogive.

Value

A matrix with one row for each value of theta and one column for each category, named 1 to m + 1.

Examples

```
prob_grm(c(-2, 0, 2))
prob_grm(c(-2, 0, 2), a = 2, b = c(-1, 1))
```

prob_grsm *Category response curves of the generalized rating scale model*

Description

A partial credit model whose step parameters are split into the difficulty of the item and thresholds shared by the items of a response scale: $\delta_v = b + \tau_v$ in prob_gpcm(). With $a = 1$ this is the rating scale model (Andrich, 1978), otherwise the generalized rating scale model (Muraki, 1990).

Usage

```
prob_grsm(theta, a = 1, b = 0, thresholds = c(-1, -0.4, 0.4, 1), D = 1)
```

Arguments

theta	Numeric vector of abilities.
a	Discrimination.
b	Difficulty (location).
thresholds	Thresholds $\tau_1, \ldots, \tau_m$.
D	Scaling constant: 1 keeps the logistic metric, and 1.702 makes the curve almost coincide with the normal ogive.

Value

A matrix with one row for each value of theta and one column for each category, named 1 to m + 1.

References

Andrich, D. (1978). A rating formulation for ordered response categories. *Psychometrika*, 43(4), 561-573.

Muraki, E. (1990). Fitting a polytomous item response model to Likert-type data. *Applied Psychological Measurement*, 14(1), 59-71.

Examples

```
prob_grsm(c(-2, 0, 2))
prob_grsm(c(-2, 0, 2), b = 1, thresholds = c(-0.5, 0.5))
```

---

prob_mirt *Response surface of a two-dimensional dichotomous item*

Description

The probability of a correct response to an item that measures two abilities. In the compensatory model it is a logistic function of a weighted sum of the abilities,

$$P(\theta_1, \theta_2) = c + \frac{d - c}{1 + \exp(-D(a_1(\theta_1 - b_1) + a_2(\theta_2 - b_2)))},$$

so that a high ability on one dimension can make up for a low ability on the other; in the non-compensatory model it is the product of one logistic function for each ability, so that a low ability on either dimension keeps the probability low.

Usage

```
prob_mirt(
  theta1,
  theta2,
  a = c(1, 1),
  b = c(0, 0),
  c = 0,
  d = 1,
  D = 1.702,
  compensatory = TRUE
)
```

Arguments

theta1, theta2	Numeric vectors of abilities on the two dimensions.
a	Discriminations on the two dimensions.
b	Locations on the two dimensions.
c	Lower asymptote.
d	Upper asymptote.
D	Scaling constant.
compensatory	TRUE for the compensatory model, FALSE for the non-compensatory one.

Value

A matrix with one row for each value of theta1 and one column for each value of theta2.

Examples

```
theta <- seq(-4, 4, length.out = 5)
prob_mirt(theta, theta)
prob_mirt(theta, theta, a = c(2, 0.5), compensatory = FALSE)
```

---

prob_nrm *Category response curves of the nominal response model*

Description

The probabilities of the unordered categories of an item, with the first category as the reference, whose exponent is 0, and each of the others with its own discrimination and difficulty (the parameterization of Baker, 1992):

$$P(X = k \mid \theta) = \frac{\exp(z_k)}{\sum_h \exp(z_h)}, \quad z_1 = 0, \quad z_k = Da_k(\theta - b_k).$$

Usage

```
prob_nrm(theta, a = c(-4, -2, 2, 4), b = c(-4, -2, 2, 4), D = 1)
```

Arguments

theta	Numeric vector of abilities.
a	Discriminations of the categories after the first.
b	Difficulties of the categories after the first.
D	Scaling constant: 1 keeps the logistic metric, and 1.702 makes the curve almost coincide with the normal ogive.

Value

A matrix with one row for each value of theta and one column for each category, named a, b, c, ...

References

Baker, F. B. (1992). *Item Response Theory: Parameter Estimation Techniques*. New York: Marcel Dekker.

Examples

```
prob_nrm(c(-2, 0, 2))
```

---

prob_tirt *Probit response functions of Thurstonian IRT*

Description

The probability of a binary outcome as the standard normal distribution function Φ of the latent trait η, in three forms:

`prob_tirt()` a single outcome, with the parameters of factor analysis, $\Phi((-\gamma + \lambda\eta)/\sqrt{\psi})$;

`prob_tirt_pair()` the choice of item i over item k, $\Phi((-\gamma + \lambda_i\eta - \lambda_k\eta)/\sqrt{\psi_i + \psi_k})$;

`prob_tirt_intercept()` the same comparison in intercept and slope form, $\Phi(\alpha + (\beta_i - \beta_k)\eta)$.

Usage

```
prob_tirt(eta, gamma = 0, lambda = 0.5, psi = 0.5)

prob_tirt_pair(
  eta,
  gamma = 0,
  lambda_i = 0.5,
  lambda_k = 0,
  psi_i = 0.5,
  psi_k = 0.5
)

prob_tirt_intercept(eta, alpha = 0, beta_i = 0.5, beta_k = 0)
```

Arguments

`eta`	Numeric vector of trait values.
`gamma`	Threshold.
`lambda`, `lambda_i`, `lambda_k`	Factor loadings.
`psi`, `psi_i`, `psi_k`	Error (unique) variances.
`alpha`	Intercept.
`beta_i`, `beta_k`	Slopes.

Value

A numeric vector of probabilities, one for each value of `eta`.

Examples

```
eta <- c(-2, 0, 2)
prob_tirt(eta)
# the pair and intercept forms agree when alpha = -gamma / sqrt(psi_i + psi_k)
# and each beta = lambda / sqrt(psi_i + psi_k)
prob_tirt_pair(eta, gamma = 0.5, lambda_i = 1, lambda_k = 0.2)
prob_tirt_intercept(eta, alpha = -0.5, beta_i = 1, beta_k = 0.2)
```

`response_pattern` *Response pattern of a raw score*

Description

The pattern in which the first `score` of `n` items are answered correctly, as the Estimation tabs of the app assume: a raw score of 2 on five items is the pattern 11000.

Usage

```
response_pattern(score, n = 5)
```

Arguments

`score`	Raw score, from 0 to `n`.
`n`	Number of items.

Value

A numeric vector of `n` ones and zeros.

Examples

```
response_pattern(2)
```

run_app *Explore IRT models and how their parameters are estimated*

Description

Launches a Shiny app that shows item response theory (IRT) with interactive plots, in two menus. The Models menu draws the item response functions of the main IRT models, so that you can see how each item parameter shapes them: the probability of each possible response is plotted against the latent trait and redrawn as soon as a parameter is changed. It covers the Rasch model and the one- to four-parameter logistic models for dichotomous items; the partial credit, generalized partial credit, graded response, rating scale, generalized rating scale and nominal response models for polytomous items; the probit response functions of Thurstonian IRT; and the response surface, contour plot and information surface of a dichotomous item that measures two abilities (multidimensional IRT). The Estimation menu shows how the estimates are found: a person's ability by maximum likelihood with the Newton-Raphson method (Rasch model), an item's discrimination and difficulty by maximum likelihood (two- and three-parameter logistic models), and a person's ability by Bayesian expected a posteriori (EAP) and maximum a posteriori (MAP) estimation. Its examples follow de Ayala (2009), and the default settings reproduce his published results. The app has a dark mode, switched by the button at the right of the navbar.

Usage

```
run_app(
  analytics = FALSE,
  keep_alive = TRUE,
  onStart = NULL,
  options = list(),
  enableBookmarking = NULL,
  uiPattern = "/",
  ...
)
```

Arguments

analytics	TRUE to add Google Analytics, through the cookie banner shared with the sites of sedzinfo.github.io, which loads it only after the visitor accepts. Meant for the public deployment (see app.R); off by default, so that running the app yourself sends nothing.
keep_alive	TRUE (the default) to ping the server every 15 seconds while the page is open, so that proxies do not close idle connections. FALSE lets the app go to sleep when nobody uses it, as app.R does on shinyapps.io, where an app that stays awake uses up the active hours of the plan.
onStart	A function that will be called before the app is actually run. See shinyApp().
options	Named options passed to runApp(), for example list(port = 8080). See shinyApp().
enableBookmarking	See shinyApp().

uiPattern — See `shinyApp()`.

... — Arguments passed to `golem::with_golem_options()`, which the app can read with `golem::get_golem_options()`.

Details

Each tab of the Models menu plots, for a single item, the probability of each possible response as a function of the person's position on the latent trait (ability), written θ (η in the TIRT tab). The basic building block is the four-parameter logistic function

$$P(X = 1 \mid \theta) = c + \frac{d - c}{1 + \exp(-Da(\theta - b))}$$

the probability of a correct (or keyed) response, $X = 1$, where a is the discrimination (how steeply the probability rises with ability), b the difficulty (the ability at which the probability is halfway between its lower and upper limits), c the lower asymptote (the probability at very low abilities, as with guessing) and d the upper asymptote (the probability at very high abilities). D is the scaling constant: 1 keeps the logistic metric, and 1.702 makes the logistic curve almost coincide with the normal ogive, the cumulative normal distribution. The Rasch-4PL tab plots this function itself, the GRM tab uses it with $c = 0$ and $d = 1$ for the boundaries between categories, and the MIRT Dichotomous tab extends it to two abilities; the other models are given in their sections.

The five tabs on a single trait, Rasch-4PL to NRM, share a layout. A Scaling Constant choice sets D to 1 (the default) or 1.702. The ability axis runs from -6 to 6, so a curve centred beyond that range, which most location sliders allow (they reach -10 and 10), moves out of view. The plots have a marker at every 0.1 of ability: hovering shows the values, clicking a legend entry hides or shows its curve, and the plot can be zoomed and panned. A horizontal gray line marks a probability of 0.5, and the vertical line at 0 is the zero line of the axis. Every plot is redrawn as soon as a control changes.

The tabs of the Estimation menu use the same function with $d = 1$ and no scaling constant ($D = 1$), and write α for the discrimination and δ for the difficulty:

$$P(u = 1 \mid \theta) = c + (1 - c)\frac{\exp(\alpha(\theta - \delta))}{1 + \exp(\alpha(\theta - \delta))}.$$

The Rasch model fixes $\alpha = 1$ and $c = 0$, the two-parameter logistic model (2PL) fixes $c = 0$, and the three-parameter logistic model (3PL) estimates all three. The likelihood of a set of responses is the product of the probabilities of the responses given, and the log-likelihood is its logarithm.

The app has eleven tabs, described in the sections below: Rasch-4PL, PCM-GPCM, GRM, RSM-GRSM, NRM, TIRT and MIRT Dichotomous in the Models menu, and MLE Rasch, MLE 2PL, MLE 3PL and EAP-MAP Rasch in the Estimation menu. The models and estimation methods are also available as functions, such as `prob_gpcm()` and `eap_map()`, and each plot of the app as a `plot_*()` function, such as `plot_category_curves()`.

Value

A Shiny app object (`shiny::shinyApp()`). Printing it, which is what happens when you call `run_app()` at the console, starts the app in the browser or the RStudio viewer; you can also pass it to `shiny::runApp()`.

Rasch-4PL

The item characteristic curve of a dichotomous item under the four-parameter logistic model (4PL) and the models nested in it, which fix some of its parameters.

- **Controls:** the scaling constant; the discrimination a (-10 to 10, default 1); the difficulty b (-10 to 10, default 0); the guessing parameter c (0 to 1, default 0); the inattentiveness parameter d (0 to 1, default 1); and the probability to plot: $P(X = 1 \mid \theta)$ alone, or $P(X \mid \theta)$ (the default), which adds the probability of an incorrect response, $P(X = 0 \mid \theta) = 1 - P(X = 1 \mid \theta)$.
- **Plot:** the chosen curves across ability, with a legend. The title names the model that the current values define: Rasch Model when $a = 1$, $c = 0$ and $d = 1$; 1PL or 2PL Model when $c = 0$ and $d = 1$ but a is not 1 (with a single item, a discrimination shared by all items, as in the 1PL, cannot be told apart from one specific to the item, as in the 2PL); 3PL Model when $c > 0$ and $d = 1$; 4PL Model (c = 0) when $c = 0$ and $d < 1$, a 4PL without guessing whose upper asymptote allows for slipping (careless errors by able respondents); and 4PL Model when $c > 0$ and $d < 1$.
- **Text:** below the controls, the name of the model followed by the values of a, b, c and d.

Raising b moves the curve to the right, so that a higher ability is needed for the same probability. The discrimination sets the steepness: at $\theta = b$ the slope is $Da(d - c)/4$, a negative a makes the probability fall as ability rises, and $a = 0$ makes the curve flat. Raising c lifts the lower end of the curve, so that even the least able answer correctly with probability c, as when guessing; lowering d pulls the upper end down, so that even the most able miss with probability $1 - d$, as through inattention. With $c = 0$ and $d = 1$ the curve crosses the gray line at $\theta = b$, where the curves of $P(X = 1 \mid \theta)$ and $P(X = 0 \mid \theta)$ also cross; in general it reaches the midpoint $(c + d)/2$ there. Choosing $D = 1.702$ makes the curve steeper by that factor.

PCM-GPCM

Category response curves of an item scored in five ordered categories, 1 to 5, under the partial credit model (PCM) and the generalized partial credit model (GPCM):

$$P(X = k \mid \theta) = \frac{\exp\left(\sum_{v=1}^{k-1} Da(\theta - \delta_v)\right)}{\sum_{h=1}^{5} \exp\left(\sum_{v=1}^{h-1} Da(\theta - \delta_v)\right)}$$

for $k = 1, \ldots, 5$, where a is the discrimination, $\delta_1, \ldots, \delta_4$ are the step parameters (one for each step from a category to the next) and an empty sum is 0, so that category 1 has the numerator 1.

- **Controls:** the scaling constant; the discrimination a (-10 to 10 in steps of 0.5, default 1); and the four step parameters (-10 to 10), with defaults -2, -0.7, 0.8 and 2 (the code asks for -0.75 and 0.75, which the sliders round to one decimal).
- **Plot:** the probabilities of the five categories, $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each. The title reads Partial Credit Model (Masters, 1982) when $a = 1$ and Generalized Partial Credit Model (Muraki, 1992) otherwise. A discrimination of 0 is rejected with a message in place of the plot.

The curves of adjacent categories k and $k+1$ cross exactly at $\theta = \delta_k$, so each step parameter moves one crossing point. With the step parameters in increasing order, each category in turn is the most likely response as ability rises; if a step parameter is lower than the one before it (reversed steps), the category between them is never the most likely. A larger a makes the curves narrower and more peaked, and a negative a reverses the order of the categories along the ability axis.

GRM

Category response curves of an item with five ordered categories under the graded response model (GRM). The model first gives, for each of the four boundaries between adjacent categories, the probability of a response in a category above the boundary,

$$P_k^*(\theta) = \frac{1}{1 + \exp(-Da(\theta - \delta_k))}, \quad k = 1, \ldots, 4,$$

and then the probability of each category as the difference between successive boundary curves, $P(X = m \mid \theta) = P_{m-1}^*(\theta) - P_m^*(\theta)$ for $m = 1, \ldots, 5$, with $P_0^* = 1$ and $P_5^* = 0$.

- **Controls:** the scaling constant; the discrimination a (0 to 10, default 1.5), shared by all boundaries; and the four difficulties, δ_1 (-2.5 to -1.5), δ_2 (-1.25 to -0.25), δ_3 (0 to 1.25) and δ_4 (1.5 to 2.5), with defaults -2, -0.8, 0.8 and 2 (the code asks for -0.75 and 0.75, which the sliders round to one decimal). The slider ranges do not overlap, which keeps the difficulties in increasing order, as the model requires: otherwise some category probabilities would be negative. For the same reason the discrimination cannot be negative: the boundary curves would then fall instead of rise and swap order, so an item that runs the other way is reverse-scored instead. At $a = 0$ the item does not depend on ability: every boundary curve is flat at 0.5, so categories 1 and 5 each get probability 0.5 and the middle categories none.
- **Plot:** $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each, titled Graded Response Model.

Each difficulty δ_k is the ability at which a response above category k becomes as likely as not. Only the first and the last can be read directly from the plot: the curve of category 1 crosses the gray line at δ_1, and that of category 5 at δ_4. Unlike the step parameters of the PCM, the difficulties are not the points where adjacent curves cross. Because the GRM gives the item a single discrimination, changing a makes all curves steeper or flatter together.

RSM-GRSM

Category response curves of an item with five ordered categories under the rating scale model (RSM) and its generalized form with a discrimination parameter (GRSM). Both are partial credit models in which each step parameter is split into the difficulty of the item and a threshold:

$$P(X = k \mid \theta) = \frac{\exp\left(\sum_{v=1}^{k-1} Da(\theta - (\delta + \tau_v))\right)}{\sum_{h=1}^{5} \exp\left(\sum_{v=1}^{h-1} Da(\theta - (\delta + \tau_v))\right)}$$

where δ is the difficulty (location) of the item and $\tau_1, \ldots, \tau_4$ are the thresholds, which in a rating scale model are the same for all items that share a response scale.

- **Controls:** the scaling constant; the discrimination a (-10 to 10, default 1); the difficulty δ (-10 to 10, default 0); and the four thresholds (-10 to 10; defaults -1, -0.4, 0.4 and 1).
- **Plot:** $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each. The title reads Rating Scale Model (Andrich, 1978) when $a = 1$ and Generalized Rating Scale Model (Muraki, 1990) otherwise. A discrimination of 0 is rejected with a message in place of the plot.

Adjacent curves k and $k + 1$ cross at $\theta = \delta + \tau_k$. Changing the difficulty shifts all five curves along the ability axis together without changing their shape, the thresholds set the spacing of the crossings around the difficulty, and the discrimination sets how peaked the curves are.

NRM

Category response curves of an item with five unordered categories, labelled a to e, under the nominal response model (NRM). Category a is the reference category, whose exponent is fixed at 0, and each of the other four categories has its own discrimination α_k (written with a Greek letter here, to avoid confusion with category a) and difficulty δ_k:

$$P(X = k \mid \theta) = \frac{\exp(z_k)}{\sum_{h=a}^{e} \exp(z_h)}, \quad z_a = 0, \quad z_k = D\alpha_k(\theta - \delta_k) \quad (k = b, \ldots, e).$$

Following Baker (1992), each exponent is written with a difficulty rather than an intercept.

- **Controls:** the scaling constant; four discriminations (-10 to 10; defaults -4, -2, 2 and 4) and four difficulties (-10 to 10; defaults -4, -2, 2 and 4). Discrimination j and difficulty j belong to the j-th category after a: 1 to b, 2 to c, 3 to d and 4 to e.
- **Plot:** $P(X = a \mid \theta)$ to $P(X = e \mid \theta)$, one coloured curve each, titled Nominal Response Model, with a note on a second, smaller line of the title that Baker's (1992) reparameterization is used.

Category k is exactly as likely as category a at $\theta = \delta_k$, because the ratio of their probabilities is $\exp(z_k)$. The discriminations order the categories along the ability axis: the category with the largest discrimination (counting 0 for category a) becomes the most likely at high abilities, the one with the smallest at low abilities, and the others can be the most likely in between. Nothing forces an order on the categories, which is what sets the NRM apart from the models for ordered categories. With the default values, categories c, a and d are in turn the most likely across the plotted range, while b and e would take over only beyond -6 and 6.

TIRT

Probit response functions of Thurstonian item response theory (TIRT), in which a binary outcome, such as the choice between two items in a forced-choice questionnaire, depends on the latent trait η through Φ, the cumulative distribution function of the standard normal distribution. The tab has three sub-tabs, each named by a formula, with these functions and controls:

- **First sub-tab:** a single binary outcome, written with the parameters of factor analysis,

  $$P(\eta) = \Phi\left(\frac{-\gamma + \lambda\eta}{\sqrt{\psi}}\right),$$

  with the threshold γ (-10 to 10, default 0), the factor loading λ (-10 to 10, default 0.5) and the error (unique) variance ψ (0 to 10, default 0.5).
- **Second sub-tab:** the choice of item i over item k in a pair,

  $$P(\eta) = \Phi\left(\frac{-\gamma + \lambda_i\eta - \lambda_k\eta}{\sqrt{\psi_i + \psi_k}}\right),$$

  with the threshold of the pair γ (-10 to 10, default 0), the loadings of the two items λ_i and λ_k (-10 to 10; defaults 0.5 and 0) and their error variances ψ_i and ψ_k (0 to 10, default 0.5 each). The formula on the sub-tab gives each item its own trait, but the plot uses the same η for both, so only the difference $\lambda_i - \lambda_k$ matters: with equal loadings the curve is flat, and the choice tells nothing about the trait.

- **Third sub-tab:** the same comparison in intercept and slope form,

$$P(\eta) = \Phi(\alpha + (\beta_i - \beta_k)\eta),$$

with the intercept α (-10 to 10, default 0) and the slopes β_i and β_k (-10 to 10; defaults 0.5 and 0). It draws the curve of the second sub-tab when $\alpha = -\gamma/\sqrt{\psi_i + \psi_k}$ and each $\beta = \lambda/\sqrt{\psi_i + \psi_k}$; with their defaults, both sub-tabs show $\Phi(0.5\eta)$.

- **Range:** each sub-tab also has a slider labelled η (1 to 100, default 6) that sets no parameter: it sets the range of the horizontal axis, from $-\eta$ to η.
- **Plot:** the probability, from 0 to 1, against η: a single curve with a marker at every 0.1, and the gray line at 0.5.

Each curve crosses 0.5 where the argument of Φ is 0, for example at $\eta = \gamma/\lambda$ in the first sub-tab. The loadings or slopes set the steepness and direction of the curve, and larger error variances make it flatter. There is no scaling constant in this tab, since the curves are normal ogives.

MIRT Dichotomous

The response surface of a dichotomous item that measures two abilities, θ_1 and θ_2 (multidimensional IRT, MIRT), on a grid of 50 by 50 points from -4 to 4 on each dimension. In the compensatory model (the default) the probability is a logistic function of a weighted sum of the two abilities,

$$P(\theta_1, \theta_2) = c + \frac{d - c}{1 + \exp(-D(a_1(\theta_1 - \delta_1) + a_2(\theta_2 - \delta_2)))},$$

so that a high ability on one dimension can make up for a low ability on the other. In the non-compensatory model it is the product of one logistic function for each ability,

$$P(\theta_1, \theta_2) = c + (d - c)\frac{1}{1 + \exp(-Da_1(\theta_1 - \delta_1))} \cdot \frac{1}{1 + \exp(-Da_2(\theta_2 - \delta_2))},$$

so that a low ability on either dimension keeps the probability low, whatever the other. Here a_1 and a_2 are the discriminations, δ_1 and δ_2 the locations on the two dimensions, and c and d the lower and upper asymptotes.

- **Controls:** a choice labelled Item Response Function, which sets the scaling constant of the logistic function: Logistic gives $D = 1$, and Normal ogive approximation (the default) gives $D = 1.702$, which makes the logistic curve almost identical to the normal ogive; whether the model is compensatory (Yes, the default, or No); the item's location on each dimension, labelled θ_1 location and θ_2 location (-10 to 10, default 0); its discrimination on each dimension, labelled θ_1 discrimination and θ_2 discrimination (-10 to 10, default 1; a value of 0 is rejected with a message); the lower asymptote c (0 to 0.4 in steps of 0.1, default 0) and the upper asymptote d (0 to 1, default 1); and optional names for the two dimensions. The axes are labelled θ_1 and θ_2, and a name typed in is shown in front of the symbol, for example Math (θ_1).
- **Item Response Surface sub-tab:** $P(\theta_1, \theta_2)$ as an interactive 3-D surface, coloured from purple (low) to yellow (high probability), with P(Y = 1) from 0 to 1 on the vertical axis; drag it to rotate it.

- **Contour Plot sub-tab:** the same surface seen from above, as contour lines of equal probability, each labelled with its probability; hovering shows the abilities and the probability under the pointer. A slider sets the approximate number of contour levels (10 to 100, default 10). Black lines mark $\theta_1 = 0$ and $\theta_2 = 0$, and both axes have the same scale. A thick line marks the contour halfway between the asymptotes, at probability $(c + d)/2$ (0.5 when $c = 0$ and $d = 1$); it is left out when that probability is not reached inside the grid, as for a very easy or very hard item. For the compensatory model a blue arrow shows the item vector, when it lies inside the grid. It starts on the line where the probability is halfway between the asymptotes, at the point of that line closest to the origin; it points in the direction in which the probability rises fastest, at the angle $\arccos(a_1/A)$ to the θ_1 axis; and its length is the multidimensional discrimination $A = \sqrt{a_1^2 + a_2^2}$. Its starting point lies at the signed distance $(a_1\delta_1 + a_2\delta_2)/A$, the multidimensional difficulty, from the origin along the direction of the arrow.
- **Information Plot sub-tab:** a 3-D surface, seen from the opposite corner, of the item information at each point of the grid: the squared slope of the probability surface, in the direction in which it rises fastest, divided by $P(1 - P)$,

$$I(\theta_1, \theta_2) = \frac{(\partial P/\partial\theta_1)^2 + (\partial P/\partial\theta_2)^2}{P(1 - P)}.$$

  For the compensatory model this is the information of the four-parameter logistic model (Magis, 2013) along the item vector,

$$I(\theta_1, \theta_2) = \frac{D^2A^2(P - c)^2(d - P)^2}{(d - c)^2P(1 - P)},$$

  which grows with the multidimensional discrimination A and, when $c = 0$ and $d = 1$, is highest along the line where $P = 0.5$. For the non-compensatory model the slope combines the two logistic parts of the model.

In the compensatory model the contour lines are straight and parallel, at right angles to the item vector, because the probability depends only on the weighted sum of the abilities; in the non-compensatory model they bend, since both abilities must be high for a high probability. Making one discrimination larger than the other turns the item vector towards the axis of that dimension, which means that the item measures mostly that ability, and moving a location shifts the surface along its dimension.

MLE Rasch

How the Newton-Raphson method finds the maximum likelihood estimate of a person's ability when the item difficulties are known. The test has five items with difficulties -1.90, -0.60, -0.25, 0.30 and 0.45 (de Ayala, 2009, p. 22), and a raw score r is taken to come from answering the first r items correctly: $r = 0$ is the response pattern 00000, $r = 2$ is 11000 and $r = 5$ is 11111.

- **Controls:** the raw score, the starting value of the estimate (-2 to 2) and the iteration to show (1 to 5).
- **Plot:** the log-likelihood of the response pattern for abilities from -6 to 6. For the chosen iteration it shows the estimate the step starts from (red dot), the tangent to the log-likelihood there (blue line) and the estimate the step ends at (green dot), given by $\theta_1 = \theta_0 - LL'(\theta_0)/LL''(\theta_0)$, with the derivatives taken from a smoothing spline of the log-likelihood. The shaded triangle shows the angle between the tangent and the horizontal axis, which is zero at the maximum.

- **Text:** the numbers of the Newton-Raphson step and the angle.

Raising the iteration shows the estimates converging on the maximum of the curve, which reproduces the person locations of de Ayala (2009, Table 2.2, p. 28). For raw scores of 0 and 5 the log-likelihood keeps rising towards minus or plus infinity, so it has no maximum and the steps move away: the maximum likelihood estimate does not exist for zero or perfect scores.

MLE 2PL

The log-likelihood surface of an item's discrimination and difficulty under the 2PL, and where its maximum lies. The app simulates the responses of 50 people, whose abilities are known and spread evenly from -3 to 3, to one item with the true discrimination and location set on the sliders, and draws new responses whenever a true value changes.

- **Controls:** the true discrimination and location (difficulty), -3 to 3.
- **Plot:** the log-likelihood of the simulated responses for every combination of discrimination and difficulty from -3 to 3 in steps of 0.1, as a 3-D surface that can be rotated and zoomed with the mouse; hovering shows the values under the pointer. The red dot marks its highest point: the maximum likelihood estimates of the two parameters. With only 50 responses they differ from the true values by sampling error, and they change with every new draw.

MLE 3PL

The same for the 3PL, to show how the guessing parameter affects the estimation. The app simulates the responses of the same 50 people from the true lower asymptote (guessing), discrimination and location set on the sliders, then computes the log-likelihood surface of discrimination and difficulty (-3 to 3 in steps of 0.15) for each of twelve assumed guessing values, 0.05, 0.10, ..., 0.60.

- **Controls:** the true lower asymptote (0.05 to 0.60), discrimination and location.
- **Plot:** the twelve surfaces drawn over each other in different colours, each with a dot at its highest point; the legend gives the guessing value of each colour, and clicking an entry hides or shows that surface. All surfaces share one vertical scale.

EAP-MAP Rasch

Bayesian estimation of a person's ability under the Rasch model, by quadrature. The test has five items with difficulties fixed at -2.155, -0.245, 0.206, 0.984 and 1.211 (de Ayala, 2009, Table 4.4), and raw scores map to response patterns as in the MLE Rasch tab. With prior weights $A(X_r)$ and likelihood $L(X_r)$ at the quadrature points X_r,

$$EAP = \frac{\sum_r X_r L(X_r) A(X_r)}{\sum_r L(X_r) A(X_r)}$$

is the mean of the posterior distribution, and MAP is the quadrature point where the posterior $L(X_r)A(X_r)$ is highest.

- **Controls:** the raw score; the number of quadrature points (10 to 100, spread evenly from -4 to 4); and the mean, standard deviation and skewness of the prior, a skew-normal distribution (`fGarch::dsnorm()`) that is normal when the skewness is 1.

- **Separate sub-tab, four plots:** the prior weights at the quadrature points; the item trace lines, the probability of each observed response across ability (blue for items answered correctly, red for items answered incorrectly); the likelihood, the product of those probabilities; and the posterior, with lines and values for the EAP (blue) and the MAP (red).
- **Combined sub-tab:** the prior (red), likelihood (blue) and posterior (black, rescaled to be comparable) in one plot, with dashed lines at the prior mean and the posterior mean.

The default settings replicate de Ayala (2009, p. 79). A narrower prior pulls the EAP towards the prior mean, more quadrature points make both estimates more precise, and, unlike the maximum likelihood estimate, EAP and MAP stay finite for raw scores of 0 and 5.

Acknowledgments

The plots of this app were originally developed by Wes Bonifay and Metin Bulus for the irtDemo package (Bulus and Bonifay, 2022), available on CRAN at https://cran.r-project.org/package=irtDemo; the title of the app, IRT Models and Estimation using IRTDemo, reflects this origin. Dimitrios Zacharatos modified and expanded them in the modelsirt and mleirt packages, which this package brings together: he consolidated the IRT models into a single interface, rebuilt the visualisations with Plotly for interactivity, added further IRT models and improved the user interface. Special thanks to Kostas Maistrelis (https://github.com/maisk) for contributing the implementation of the Thurstonian IRT curves. The tabs keep the credits shown in the apps: the Rasch-4PL, PCM-GPCM, RSM-GRSM and MLE 3PL tabs are credited to Metin Bulus, the EAP-MAP Rasch tab to Metin Bulus and Wes Bonifay, and the GRM, NRM, MIRT Dichotomous, MLE Rasch and MLE 2PL tabs to an unknown contributor.

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Examples

```
if (interactive()) run_app()

# on a fixed port
if (interactive()) run_app(options = list(port = 8080))
```

`simulate_responses` *Simulate dichotomous responses*

Description

Draws one response of each person to an item, correct with the probability given by `prob_4pl()` (with $D = 1$ and $d = 1$).

Usage

```
simulate_responses(theta, a = 1, b = 0, c = 0)
```

Arguments

`theta`	Numeric vector of the abilities of the people.
`a`	Discrimination of the item.
`b`	Difficulty (location) of the item.
`c`	Lower asymptote of the item.

Value

An integer vector of ones (correct) and zeros (incorrect), one for each value of `theta`.

Examples

```
set.seed(1)
simulate_responses(seq(-3, 3, length.out = 10), a = 1, b = -0.5)
```

surface_maximum *Highest point of a log-likelihood surface*

Description

The maximum likelihood estimates of discrimination and difficulty on the grid of a surface from loglik_surface().

Usage

```
surface_maximum(loglik, a = attr(loglik, "a"), b = attr(loglik, "b"))
```

Arguments

loglik	Matrix of log-likelihoods, one row for each value of a and one column for each value of b.
a, b	The grids of discriminations and difficulties of the surface; by default those that loglik_surface() keeps with it.

Value

A list with a, b and loglik, the discrimination, difficulty and log-likelihood at the highest point; at a tie, the first in column order.

Examples

```
set.seed(1)
theta <- seq(-3, 3, length.out = 50)
surface <- loglik_surface(simulate_responses(theta, a = 1, b = -0.5), theta)
surface_maximum(surface)
```

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